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Submitted by

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DECLARATION

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ABSTRACT

The rapid growth of user-generated textual content across social media platforms, online forums, and digital communication channels has created a strong demand for intelligent systems capable of understanding human emotions. Unlike traditional sentiment analysis, which categorizes text into broad classes such as positive, negative, or neutral, emotion detection aims to identify fine-grained emotional states. This task becomes more complex in real-world scenarios where a single sentence may express multiple emotions simultaneously, making it a multi-label classification problem. Additionally, challenges such as contextual ambiguity, sarcasm, and severe class imbalance further complicate accurate emotion recognition. This study presents a transformer-based approach for multi-label emotion classification using a fine-tuned RoBERTa model on the GoEmotions dataset, which contains approximately 121,000 Reddit comments annotated across 27 emotion categories. The proposed system leverages deep contextual embeddings generated by the RoBERTa architecture to capture semantic relationships and nuanced emotional cues within text. Unlike traditional machine learning methods such as Support Vector Machines and Naïve Bayes, which rely on handcrafted features, transformer-based models utilize self-attention mechanisms to learn rich contextual representations, significantly improving performance in complex natural language processing tasks. The methodology involves several key stages, including data preprocessing, tokenization, model training, and evaluation. Initially, irrelevant columns are removed, and the dataset is cleaned and transformed into a multi-label format using multi-hot encoding. The RoBERTa tokenizer is employed to convert text into subword tokens, ensuring efficient handling of rare and unseen words. These tokens are then passed through the RoBERTa encoder, which captures bidirectional contextual information using self-attention. A fully connected classification layer with sigmoid activation is applied to generate independent probabilities for each emotion label, enabling multi-label predictions. One of the primary contributions of this work is addressing the issue of class imbalance, which is prevalent in the GoEmotions dataset. To mitigate this, a weighted Binary Cross-Entropy (BCE) loss function is implemented, where class weights are computed based on label distribution. This approach penalizes misclassification of minority classes more heavily, thereby improving the model's ability to detect less frequent emotions. Additionally, instead of using a fixed decision threshold, an adaptive threshold tuning strategy is applied. By evaluating multiple threshold values and selecting the one that maximizes the F1-score, the model achieves a better balance between precision and recall.

The model is trained using the Adam optimizer with a learning rate of 2e-5 and weight decay for regularization. Performance is evaluated using standard multi-label metrics, including micro and macro F1-score, precision, and recall. Experimental results demonstrate that the proposed model achieves a micro F1-score of approximately 0.44–0.48, with a balanced trade-off between precision (around 42%) and recall (around 57%). The model performs particularly well on frequently occurring emotions such as joy, anger, and sadness, while maintaining reasonable performance on minority classes due to the imbalance-aware optimization strategy. Comparative analysis with baseline models, including traditional machine learning approaches and earlier deep learning architectures such as CNNs and BERT, indicates that the RoBERTa- based model provides superior contextual understanding and overall classification performance. The training loss curve further confirms effective learning and convergence, with minimal overfitting observed during training. Despite its effectiveness, the model has certain limitations, including high computational requirements due to the use of a large transformer model and challenges in accurately detecting highly ambiguous or sarcastic expressions. Future work may focus on improving computational efficiency through lightweight architectures, enhancing generalization across domains, and incorporating advanced techniques such as attention refinement or domain adaptation. In conclusion, this research demonstrates that transformer-based models, combined with imbalance-aware optimization and adaptive threshold tuning, provide a robust and scalable solution for multi-label emotion detection. The proposed approach contributes to the advancement of emotion-aware natural language processing systems, with potential applications in sentiment analysis, conversational AI, mental health monitoring, and human- computer interaction

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CHAPTER 1

1.1 Introduction

The rapid expansion of user-generated content on social media platforms, online forums, and digital communication systems has created a growing need for intelligent methods to analyse human emotions from text. Unlike traditional sentiment analysis, which classifies text into simple categories such as positive, negative, or neutral, emotion detection focuses on identifying fine-grained emotional states like joy, anger, sadness, and fear. This task becomes more challenging in real-world scenarios where a single piece of text may express multiple emotions simultaneously, making it a multi-label classification problem. Additional challenges such as contextual ambiguity, sarcasm, and class imbalance further complicate accurate emotion recognition. With the advancement of deep learning, transformer-based models like RoBERTa have significantly improved natural language understanding through contextual embeddings and self-attention mechanisms. This study leverages these capabilities to develop an efficient multi-label emotion classification system using the GoEmotions dataset.

1.2 MOTIVATION BEHIND PROJECT TOPIC

The rapid growth of online text data demands accurate emotion understanding beyond basic sentiment analysis. Traditional methods struggle with multi-label emotions and class imbalance. This project is motivated by the need for a robust system using transformer models like RoBERTa to improve emotion detection accuracy in real-world applications.

1.3 AIM AND OBJECTIVE OF THE WORK

The primary aim of this work is to develop an efficient and scalable multi-label emotion classification system capable of accurately identifying multiple co-occurring emotions within textual data. By leveraging transformer-based architectures, the study focuses on improving the understanding of contextual and semantic relationships in language, thereby enhancing emotion detection performance in real-world scenarios. To achieve this aim, several key objectives are defined. First, to preprocess and structure the GoEmotions dataset by cleaning noise, removing irrelevant attributes, and converting emotion labels into a multi- label format suitable for model training. Second, to implement a robust text representation mechanism using the RoBERTa tokenizer and encoder, enabling the extraction of deep contextual embeddings from input text. Third, to design a multi-label classification framework using a sigmoid activation function, allowing the model to predict multiple emotions simultaneously.

Another important objective is to address the issue of class imbalance by incorporating a weighted Binary Cross-Entropy loss function, ensuring better learning for minority emotion classes. Additionally, the study aims to optimize prediction performance through adaptive threshold tuning to balance precision and recall effectively. Finally, the model is evaluated using standard metrics such as precision, recall, and F1-score to assess its effectiveness and generalization capability across diverse emotional categories.

CHAPTER 2

Literature Survey

Emotion detection from text has evolved significantly over the years, transitioning from traditional machine learning techniques to advanced deep learning and transformer-based approaches. Early research primarily relied on lexicon-based methods and classical machine learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Decision Trees. These methods used handcrafted features like Bag-of-Words and TF-IDF representations to classify emotions. Although these approaches were simple and interpretable, they lacked contextual understanding and were ineffective in handling complex linguistic patterns such as sarcasm, negation, and implicit emotional expressions.

With the advancement of deep learning, models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, improved emotion detection performance by automatically learning hierarchical features from text. However, these models still struggled with capturing long-range dependencies and contextual relationships in sentences. To address these limitations, hybrid architectures such as BERT-CNN were proposed, combining transformer-based embeddings with CNN feature extraction to enhance emotion classification performance.

A major breakthrough in natural language processing came with the introduction of transformer architectures by Vaswani et al. (2017), which utilized self-attention mechanisms to model contextual relationships more effectively. Building on this, Jacob Devlin et al. (2019) introduced BERT, a bidirectional transformer model that significantly improved performance across various NLP tasks, including emotion detection. Transformer-based models generate deep contextual embeddings, allowing better understanding of semantic relationships and emotional nuances in text.

Further advancements were made with RoBERTa, an optimized version of BERT developed by Liu et al. (2019), which improved pre-training strategies and achieved superior performance in language understanding tasks. Studies have shown that RoBERTa consistently outperforms traditional models and even BERT in emotion classification tasks due to its robust contextual learning capabilities. Comparative analyses also demonstrate that transformer-based models outperform

classical machine learning and recurrent models in both accuracy and generalization ability. The introduction of the GoEmotions dataset by Dorottya Demszky et al. (2020) marked a significant milestone in emotion detection research. This dataset contains fine-grained annotations of 27 emotion categories, enabling multi-label emotion classification and providing a benchmark for evaluating modern models. However, the dataset presents challenges such as class imbalance and overlapping emotion labels, which affect model performance. Recent studies have focused on improving transformer-based models for multi-label emotion classification. Harutyunyan and Kumar (2026) compared classical machine learning, BiLSTM, and transformer models on the GoEmotions dataset, concluding that transformer models provide better performance, especially in handling complex and rare emotions. Similarly, research on fine-tuning BERT for emotion detection demonstrates significant improvements over traditional models, achieving higher accuracy and F1-score.

To further enhance performance, researchers have introduced modifications to transformer architectures. For instance, Alqarni et al. (2025) proposed an Emotion-Aware RoBERTa model that incorporates an emotion-specific attention mechanism and TF-IDF-based gating to focus on emotionally relevant tokens. This approach improves the detection of subtle and minority emotions by dynamically adjusting attention weights. Additionally, ensemble methods combining multiple transformer models have been explored to boost prediction accuracy and robustness.

Despite these advancements, challenges such as class imbalance, ambiguity in emotional expressions, and domain generalization remain open research problems. Recent surveys highlight the importance of transformer-based architectures and multimodal approaches in advancing emotion recognition systems, emphasizing their potential in real-world applications such as mental health monitoring, conversational AI, and human-computer interaction.

In summary, the literature demonstrates a clear progression from traditional machine learning to advanced transformer-based models, with RoBERTa emerging as a highly effective solution for multi-label emotion classification. However, ongoing research continues to focus on improving model robustness, efficiency, and generalization capabilities.

Sr. No	Paper Allocation	Author & Year	Methods
1.	GoEmotions: A Dataset of Fine-Grained Emotions	Demszky et al., 2020	Introduced a large-scale dataset with 27

			emotion labels and applied BERT for multi-label emotion classification.
2.	BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding	Devlin et al., 2019	Proposed BERT using bidirectional transformer architecture with masked language modelling for contextual representation.
3.	Attention Is All You Need	Vaswani et al., 2017	Introduced transformer the architecture with self-attention mechanism for capturing long-range dependencies in text.
4.	RoBERTa: A Robustly Optimized BERT Pretraining Approach	Liu et al., 2019	Improved BERT training strategy with dynamic masking and larger training data for better performance.
5.	Emotion-Specific Attention Mechanism with RoBERTa	Alqarni et al., 2025	Enhanced RoBERTa with emotion-aware attention and TF-IDF gating to focus on emotionally relevant words.

Table 1 – Literature Survey

CHAPTER 3

Software Requirement Specification

3.1 INTRODUCTION

The Software Requirement Specification (SRS) defines the functional and non-functional requirements of the proposed multi-label emotion detection system. It provides a detailed description of system features, performance expectations, and constraints to ensure clarity during development. This document outlines how the system processes textual data, applies transformer-based models like RoBERTa, and generates emotion predictions. It serves as a blueprint for developers and stakeholders, ensuring that the system meets intended objectives efficiently and reliably.

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3.1.1 Project Purpose

The purpose of this project is to develop a multi-label emotion detection system using the RoBERTa model to accurately identify multiple emotions in text. It aims to improve contextual understanding and overcome limitations of traditional sentiment analysis for real-world applications.

3.1.2 Project Scope

The scope includes building and evaluating a text-based emotion classification system using the GoEmotions dataset. It covers preprocessing, model training, and performance evaluation, focusing only on English text data, with future scope for real-time and multilingual extensions.

3.1.3 User Classes and Characteristics

The system is intended for researchers, developers, and analysts working in NLP and data science. Users are expected to have basic knowledge of machine learning and programming. End users may include organizations analysing social media or customer feedback for emotional insights and decision-making.

3.1.4 Assumptions and Dependencies

•Assumptions:

1.Input text is in English language.

2.Users have basic technical knowledge of the system.

3.The dataset used is properly labelled and pre-processed.

4.Required libraries and hardware support (CPU/MPS/GPU) are available.

•Dependencies:

1.Depends on libraries like PyTorch, Transformers, and Scikit-learn.

2.Requires availability of the GoEmotions dataset.

3.Performance depends on computational resources.

4.Relies on pre-trained RoBERTa model availability.

3.2 FUNCTIONAL REQUIREMENT

3.2.1 System Feature 1(Functional Requirement)

Description:

The system shall accept textual input from the user or dataset and perform necessary preprocessing before feedingit into the model.

Functional Requirements:

- The system shall accept raw text input from a dataset or user interface.
- The system shall clean and preprocess text by removing unnecessary columns and handling missing values.
- The system shall convert text into a structured format suitable for tokenization.
- The system shall prepare multi-label emotion vectors corresponding to each text input.

Inputs: Raw textual data (sentences/comments)

Outputs: Cleaned and structured text data ready for tokenization

3.3 NON-FUNCTIONAL REQUIREMENT

3.3.1 Performance Requirements

1. The system should process and classify text inputs efficiently with minimal latency.

2. Model training should be completed within a reasonable time based on available hardware (CPU/GPU/MPS).

3. The system should handle large datasets (e.g., 100k+ samples) without significant performance degradation.

4. Prediction time for new inputs should be fast enough for near real-time applications.

5. The system should maintain stable performance with consistent accuracy and minimal resource consumption.

3.3.2 Safety Requirements:

- The system shall ensure that user data is handled securely and not exposed to unauthorized access.
- The system shall avoid generating harmful, biased, or misleading emotion predictions.
- Proper validation should be implemented to prevent invalid or malicious input.
- The system should include error handling to prevent crashes during execution.
- Sensitive data used for training or testing must be protected and handled according to data privacy guidelines.

3.3.3 Security Requirements

The system shall ensure secure handling of data through proper access control and protection mechanisms. It must prevent unauthorized access, safeguard stored datasets and model outputs, and validate inputs to avoid malicious usage. Additionally, it should maintain data confidentiality and integrity throughout processing and storage.

3.3.4 Software Quality Attributes

1. Runtime System Qualities: The system should operate reliably during execution without crashes or failures. It must efficiently manage memory and computational resources, ensuring stable performance across different environments.

2. Functionality: The system must accurately perform multi-label emotion classification from textual input. It should correctly preprocess data, apply the model, and generate meaningful emotion predictions.

3. Performance:
The system should provide fast processing for both training and inference. It must handle large datasets efficiently while maintaining acceptable response times for predictions.

4. Security:
The system should protect data from unauthorized access and ensure secure handling of inputs and outputs. It must prevent misuse and maintain confidentiality and integrity of information.

5. Availability:
The system should be consistently accessible and operational when required. It must minimize downtime and ensure continuous service for users.

6. Usability:
The system should be user-friendly and easy to operate, even for users with basic technical knowledge. Clear outputs and simple interaction should enhance user experience.

7. Interoperability:
The system should integrate smoothly with other applications, tools, or platforms. It must support compatibility with common libraries and frameworks used in machine learning and NLP.

3.4 SYSTEM REQUIREMENT

3.4.1 Software Requirements

Your paragraph text

- Operating System: Windows, macOS, or Linux
- Programming Language: Python (version 3.8 or above)
- Libraries/Frameworks: PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas
- Development Tools: Jupyter Notebook / VS Code / PyCharm
- Model Requirements: Pre-trained RoBERTa model (Roberta-large or Roberta-base)
- Dataset: GoEmotions dataset (CSV format)
- Package Manager: pip or Conda for dependency management

3.4.2. HARDWARE REQUIREMENTS

1. Processor:Inteli5/AMDRyzen 5 or higher
2. RAM: Minimum8GB(16GB recommended for training large models)
3. Storage:Atleast10–20GBfree space for dataset and model files
4. GPU (Optional): NVIDIAGPU with CUDA support or Apple MPS for faster training
5. System Type:64-bitarchitecture

3.4.3. Functional Requirements

The system shall accept textual input and preprocess it for analysis. It shall tokenize text using a transformer tokenizer and perform multi-label emotion classification using a trained model. The system must generate probabilities for each emotion and convert them into predictions using thresholding. It shall also evaluate performance using metrics like precision, recall, and F1-score.

3.4.4. Non-Functional Requirements

The system should provide efficient performance with minimal processing time for predictions. It must ensure reliability and stability during training and inference. The system should handle large datasets without significant degradation. It must maintain data security and integrity, be user-friendly, and support scalability for future enhancements and integration with other applications.

CHAPTER 4

DESIGN AND MODELLING OF SYSTEM

Design and Modelling of System:

The design and modelling of the proposed multi-label emotion detection system focus on creating an efficient, scalable, and accurate framework capable of identifying multiple emotions from textual data. The system leverages transformer-based deep learning techniques to process input text, extract contextual features, and generate emotion predictions. The overall architecture is modular, ensuring flexibility, maintainability, and ease of future enhancements.

System Architecture Overview:

The system follows a pipeline-based architecture consisting of multiple stages: data input, preprocessing, tokenization, model training, prediction, and evaluation. Each component is designed to perform a specific function, contributing to the overall workflow. The architecture ensures smooth data flow from raw input to final emotion classification output. At a high level, the system takes textual data as input, processes it through a trained model, and outputs a set of predicted emotions. The modular design allows individual components to be improved or replaced without affecting the entire system.

Data Input and Preprocessing Module:

The first stage of the system involves dataacquisition and preprocessing. The input is taken from the GoEmotions dataset, which contains labeled textual data. Preprocessing includes removing irrelevant columns, handling missing values, and organizing the data into a structured format. The text is cleaned to ensure consistency and reduce noise, which improves model performance. Emotion labels are converted into multi-hot encoded vectors, enabling the system to handle multiple emotions for asingle input. This step is crucial for preparing the dataset for multi-label classification.

Tokenization and Input Representation:

After preprocessing, the textual data is converted into a format suitable for the model using the RoBERTa Tokenizer. Tokenization breaks down text into subword units, allowing the model to handle rare and unknown words effectively. Each input sequence is transformed into input IDs and attention masks. Padding and truncation are applied to ensure uniform sequence length (maximum length of 100 tokens). This standardized representation allows efficient batch processing during training and inference.

Model Design (Transformer-Based Classification):

The core of the system is built using the RoBERTa model. This transformer-based architecture uses self-attention mechanisms to capture contextual relationships between words in a sentence. The model consists of an encoder that generates contextual embeddings, followed by a fully connected classification layer. A sigmoid activation function is applied to produce independent probability scores for each emotion label. This design enables multi-label classification, where multiple emotions can be predicted simultaneously. The use of RoBERTa enhances the system's ability to understand complex language patterns, including sarcasm and implicit emotions.

Loss Function and Optimization Strategy:

To address the challenge of class imbalance,thesystem employs a weighted Binary Cross-Entropy (BCE) loss function. The weights are calculatedbased on the frequency of each emotion label, ensuring that minority classes are given higher importanceduring training. The model is optimized using the Adam optimizerwith a low learning rate, which helps in stable fine- tuning of pre-trained weights. Weight decayisapplied as a regularization technique to prevent overfitting. This combination of strategies improvesthe model's generalization capability.

Training and Evaluation Module:

The training process involves feeding batches of tokenized data into the model and updating weights based on the computed loss. The dataset is split into training and testing sets (80:20 ratio) to evaluate performance. The system uses evaluation metrics such as precision, recall, and F1-score (micro and macro) to assess performance. These metrics provide a comprehensive understanding of the model's effectiveness, especially in handling imbalanced multi-label data. Training is performed over multiple epochs, and the best model is selected based on evaluation performance.

Threshold Optimization Module:

A key component of the system is threshold tuning. Since the model outputs probabilities, a threshold is required to convert these probabilities into binary predictions. Instead of using a fixed threshold, the system evaluates multiple threshold values and selects the one that maximizes the F1-score. This adaptive approach improves the balance between precision and recall, leading to better overall performance. Threshold optimization is particularly important in multi-label classification tasks.

Output and Prediction Module:

Once the model is trained, it canbe used to predict emotions for new text inputs. The system processes the input through the same pipeline—tokenization, model inference, and thresholding—to generate predicted emotion labels. The output is presented as a set of emotions associated with the input text, along with their respective probabilities. This information can be used in various applications such as sentiment analysis, chatbots, and mental health monitoring.

CHAPTER 5

ALGORITHM AND METHODS

This section describes the algorithms and methods used in the proposed multi-label emotion detection system. The approach combines data preprocessing, transformer-based modelling, imbalance-aware optimization, and threshold tuning to achieve accurate emotion classification. Each stage of the system is explained step-by-step for clarity.

Data Preprocessing Algorithm

The preprocessing stage ensures that raw textual data is clean, structured, and suitable for modeltraining.

Steps:

1. Load the dataset from a CSV file containing text and emotion labels.
2. Shuffle the dataset randomly to remove any ordering bias.
3. Remove unnecessary columns such as IDs, author names, and metadata.
4. Handle missing or invalid values by converting them into numerical form.
5. Separate input text and corresponding emotion labels.
6. Convert emotion labels into multi-label format (multi-hot encoding).
7. Store processed text and labels for further stages.

Thisstep improves data quality and ensures consistency across the dataset.

Data Splitting Algorithm

To evaluate the model effectively, the dataset is divided into training and testing sets.

Steps:

1. Take the pre-processed dataset.
2. Define the split ratio (e.g., 80% training and 20% testing).
3. Randomly divide the dataset into training and testing subsets.
4. Ensure both subsets maintain a similar distribution of emotion labels.

5. Store the split data for model training and evaluation.

This step helps measure the generalization capability of the model.

Tokenization Algorithm

Tokenization converts text into numerical form that can be processed by the model.
Steps:

1. Initialize the tokenizer using RoBERTa Tokenizer.
2. Input the cleaned text data.
3. Break each sentence into sub word tokens.
4. Convert tokens into numerical IDs.
5. Generate attention masks to differentiate real tokens from padding.
6. Apply padding and truncation to ensure a fixed sequence length (e.g., 100 tokens).
7. Store tokenized outputs for model input.

Thisstep ensures uniform input representation for efficient processing.

Model Initialization Algorithm

The model is initialized using a pre-trained transformer architecture.
Steps:

1. Load the pre-trained RoBERTa model.
2. Specify the number of output labels (equal to emotion categories).
3. Set the problem type as multi-label classification.
4. Move the model to the appropriate device (CPU/GPU/MPS).
5. Initialize model parameters for fine-tuning.

Thisstep enables the model to leverage pre-trained knowledge for improved performance.

Loss Function and Weight Calculation Algorithm

To handle class imbalance, weighted loss is used.
Steps:

1. Calculate the frequency of each emotion label in the training dataset.
2. Compute class weights using the formula:
   weight = (total samples – positive samples) / positive samples
3. Convert weights into tensor format.
4. Initialize Binary Cross-Entropy loss with logits using these weights.
5. Apply the loss function during training to penalize misclassification of minority classes.

This method ensures balanced learning across all emotion categories.

Model Training Algorithm

The training process updates model parameters to minimize loss.
Steps:

1. Initialize training parameters such as learning rate, batch size, and number of epochs.
2. Load training data in batches.
3. For each batch:
    - Pass input tokens through the model.
    - Obtain output logits.
    - Compute loss using weighted BCE loss.
    - Perform backpropagation to calculate gradients.
    - Update model weights using the Adam optimizer.
4. Repeat for all batches in each epoch.
5. Evaluate model performance after each epoch.
6. Save the best-performing model.

This iterative process enables the model to learn patterns in the data.

Evaluation Metrics Algorithm

The model performance is measured using standard multi-label metrics.
Steps:

1. Obtain model predictions (logits) for test data.
2. Apply sigmoid function to convert logits into probabilities.
3. Convert probabilities into binary predictions using a threshold.
4. Compare predicted labels with actual labels.
5. Compute evaluation metrics:
    - Precision = True Positives / (True Positives + False Positives)
    - Recall = True Positives / (True Positives + False Negatives)
    - F1-score = Harmonic mean of precision and recall
6. Calculate both micro and macro averages.
7. Store and display results.

These metrics provide a comprehensive evaluation of model performance.

Threshold Tuning Algorithm

Threshold tuning improves prediction accuracy in multi-label classification.
Steps:

1. Obtain predicted probabilities from the model.
2. Define a range of threshold values (e.g., 0.2 to 0.6).
3. For each threshold value:
    - Convert probabilities into binary predictions.

- Compute F1-score for predictions.

4. Compare F1-scores across all thresholds.
5. Select the threshold with the highest F1-score.
6. Use the optimal threshold for final predictions.

This adaptive approach improves the balance between precision and recall.

Prediction Algorithm

The system predicts emotions for new input text.

Steps:

1. Accept new textual input from the user.
2. Preprocess and clean the input text.
3. Tokenize the input using the tokenizer.
4. Pass tokenized input through the trained model.
5. Obtain probability scores using sigmoid activation.
6. Apply the optimal threshold to generate binary predictions.
7. Map predictions to corresponding emotion labels.
8. Display predicted emotions as output.

Thisalgorithm enables real-time emotion detection.

Overall Methodology Summary

The complete method integratesall individual algorithms into a unified workflow:

1. Data collection and preprocessing
2. Dataset splitting
3. Tokenization and input preparation
4. Model initialization using RoBERTa
5. Training with weighted BCE loss
6. Evaluation using precision, recall, and F1-score
7. Threshold optimization
8. Final prediction and output generation

The algorithms and methods used in this system provide a structured and efficient approach to multi-label emotion detection. By combining transformer-based modelling with imbalance- aware optimization and adaptive thresholding, the system achieves robust performance. The step-by-step design ensures clarity, scalability, and ease of implementation, making it suitable for real-world applications in natural language processing.

CHAPTER 6

Implementation and Testing

6.1 Data Acquisition

Fig 6.1.1 loading dataset

```
df = pd.read_csv("go_emotions_dataset.csv")
print("Dataset Loaded:")
print(df.head())
```

Fig 6.1.2 sampling data

```
df = df.sample(n=80000, random_state=42).reset_index(drop=True)
print("\nSampled Data (80K):")
print(df.shape)
```

Fig 6.1.3 removing unnecessary columns

```
df = df.drop(columns=[
    'id','author','subreddit',
    'link_id','parent_id',
    'created_utc','rater_id'
], errors='ignore')

print("\nColumns After Cleaning:")
print(df.columns)
```

Fig 6.1.4 preparing labels

```
label_cols = df.columns.drop(['text'])

labels = df[label_cols].apply(pd.to_numeric, errors='coerce') \
                .fillna(0) \
                .astype(np.float32)

print("\nLabels Prepared:")
print(labels.head())
```

6.2 Data Pre-Processing

```
X_train, X_test, y_train, y_test = train_test_split(
    df['text'], labels, test_size=0.2, random_state=42
)
```

Fig 6.2.1 Data splitting

```
tokenizer = RobertaTokenizer.from_pretrained('roberta-base')

def tokenize(texts):
    return tokenizer(
        list(texts),
        padding=True,
        truncation=True,
        max_length=100,
        return_tensors="pt"
    )

train_encodings = tokenize(X_train)
test_encodings = tokenize(X_test)
```

Fig 6.2.2 tokenization

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=4,
    per_device_eval_batch_size=4,
    num_train_epochs=4,
    evaluation_strategy="epoch",
    save_strategy="epoch",
    logging_dir="./logs",
    load_best_model_at_end=True,
    weight_decay=0.01,
)
```

Fig 6.2.3 Model Training

6.3 Model Testing

The trained RoBERTa model is tested using the test dataset to evaluate its ability to predict emotions for unseen text data. The testing process involves passing input text through the tokenizer and the trained model to obtain prediction probabilities for each emotion class.

The model outputs probability scores using a sigmoid activation function, as this is a multi-label classification problem. These probabilities are then processed using a threshold value to determine the most relevant emotion. From the predicted probabilities, the emotion with the highest confidence score is considered as the dominant emotion expressed in the text.

During testing, the model is able to analyse complex sentences and identify the strongest emotional tone present in the input. This simplifies interpretation while still leveraging the multi-label learning capability of the model.

Sample Prediction Output

Input: *"I feel happy but also nervous"*
Output: { joy: 0.61 }

This example shows that the model identifies the most prominent emotion in the sentence, which is useful for applications where a single dominant emotion is required.

Overall, the testing phase confirms that the model is capable of generating meaningful and interpretable predictions on unseen data.

Fig 6.3 Model prediction

CHAPTER 7

7.1 Evaluation Metrices

Evaluation metrics are essential for measuring the performance of a multi-label emotion classification system. Since a single text can have multiple emotion labels, traditional accuracy alone is not sufficient. Therefore, metrics like precision, recall, and F1-score are used to provide a more reliable evaluation.

1. Precision

Precision measures how many of the predicted emotion labels are actually correct. It indicates the model's ability to avoid false positives.

Formula:

Precision=TP+FP/TP

W her e:

- TP=True Positives
- FP=False Positives

Highprecisionmeans the model makes fewer incorrect emotion predictions.

2. Recall

Recall measures how many of the actual emotion labels are correctly identified by the model. Itreflects the model's ability to capture all relevant emotions.

Formula:

$$Recall=TP/TP+FN$$

W her e:

- FN=False Negatives

Highrecallindicates the model successfully detects most of the true emotions.

3.F1-Score

F1-scoreistheharmonic mean of precision and recall. It provides a balanced measure ofthemodel'sperformance.

Formula:

$$F1\text{-}Score=2\times Precision*Recall/Precision+Recall$$

AhigherF1-score indicates better overall performance.

4.Micro-Averaged Metrics

Micro-averagingcalculates metrics globally by considering all labels together. It is useful when dealing with imbalanced datasets.

- Focuses on overall performance
- Gives more importance to frequent classes

5. Macro-Averaged Metrics

Macro-averaging calculates metrics independently for each label and then takes the average.

- Treats all classes equally
- Useful for evaluating performance on minority classes

In this project, precision, recall, and F1-score (both micro and macro) are used to evaluate the model. These metrics provide a comprehensive understanding of how well the system performs in detecting multiple emotions, especially in the presence of class imbalance and overlapping labels.

7.2 Performance Results

The performance of the proposed multi-label emotion classification system is evaluated using standard metrics such as precision, recall, and F1-score. These metrics provide a comprehensive understanding of the model's effectiveness, especially in handling imbalanced and multi-label data.

Metric	Value
F1Score(Micro)	0.48
F1Score(Macro)	0.36
Precision	0.42
Recall	0.57

Table- 2: Performance Metrics

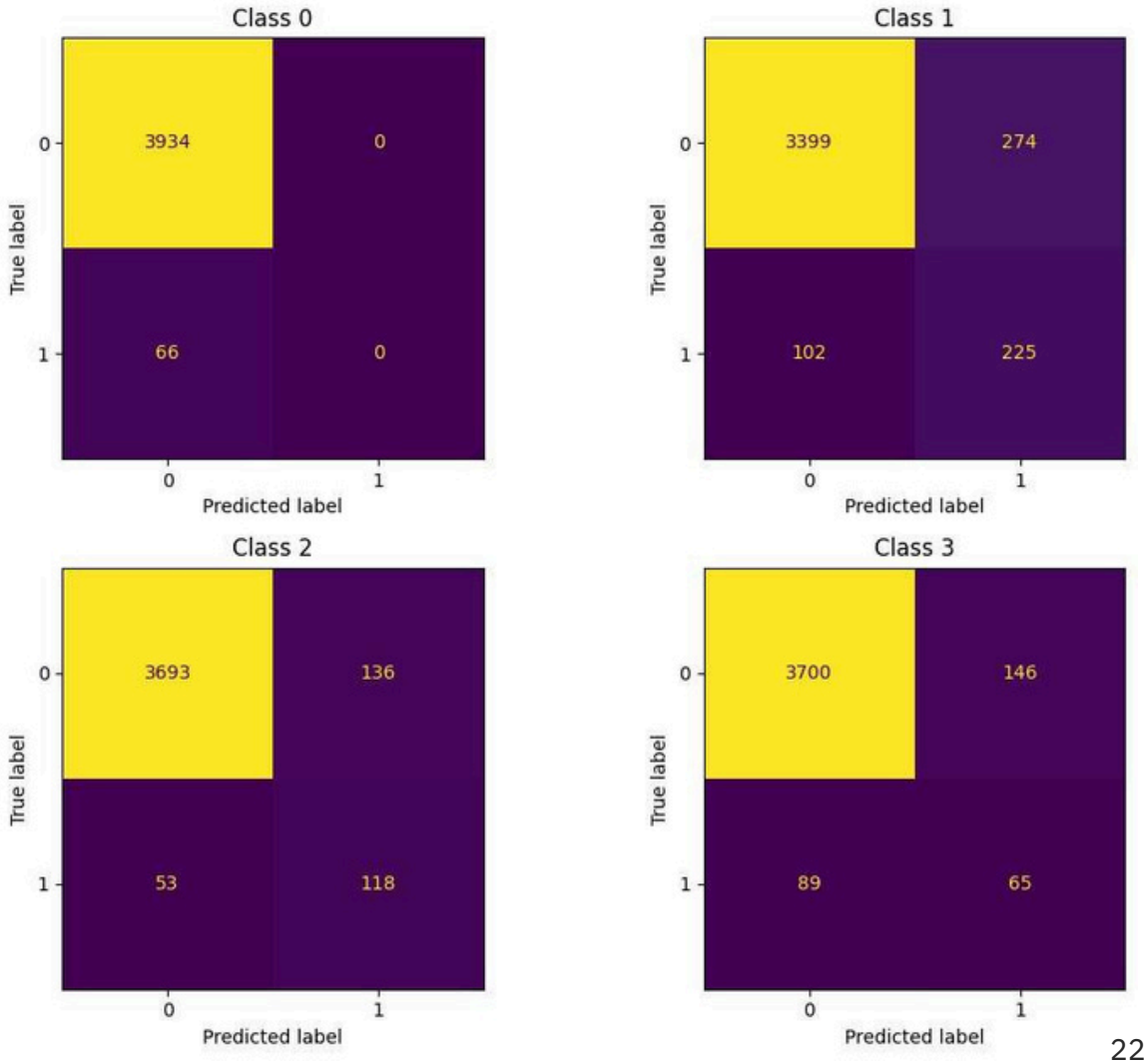
The results indicate that the model achieves moderate performance on the multi-label emotion classification task. The micro F1-score of 0.48 reflects the model's ability to perform well overall across all predictions, while the macro F1-score of 0.36 highlights challenges in predicting minority emotion classes. The precision value of 0.42 suggests that some predicted emotions may include false positives, whereas the higher recall of 0.57 indicates that the model is effective in identifying a larger proportion of actual emotions present in the text. This shows that the model prioritizes capturing relevant emotions, even at the cost of some incorrect predictions.

Overall, the results demonstrate a balanced trade-off between precision and recall, making

the model suitable for real-world applications where identifying most relevant emotions is important. However, further improvements can be made to enhance precision and performance on less frequent emotion classes.

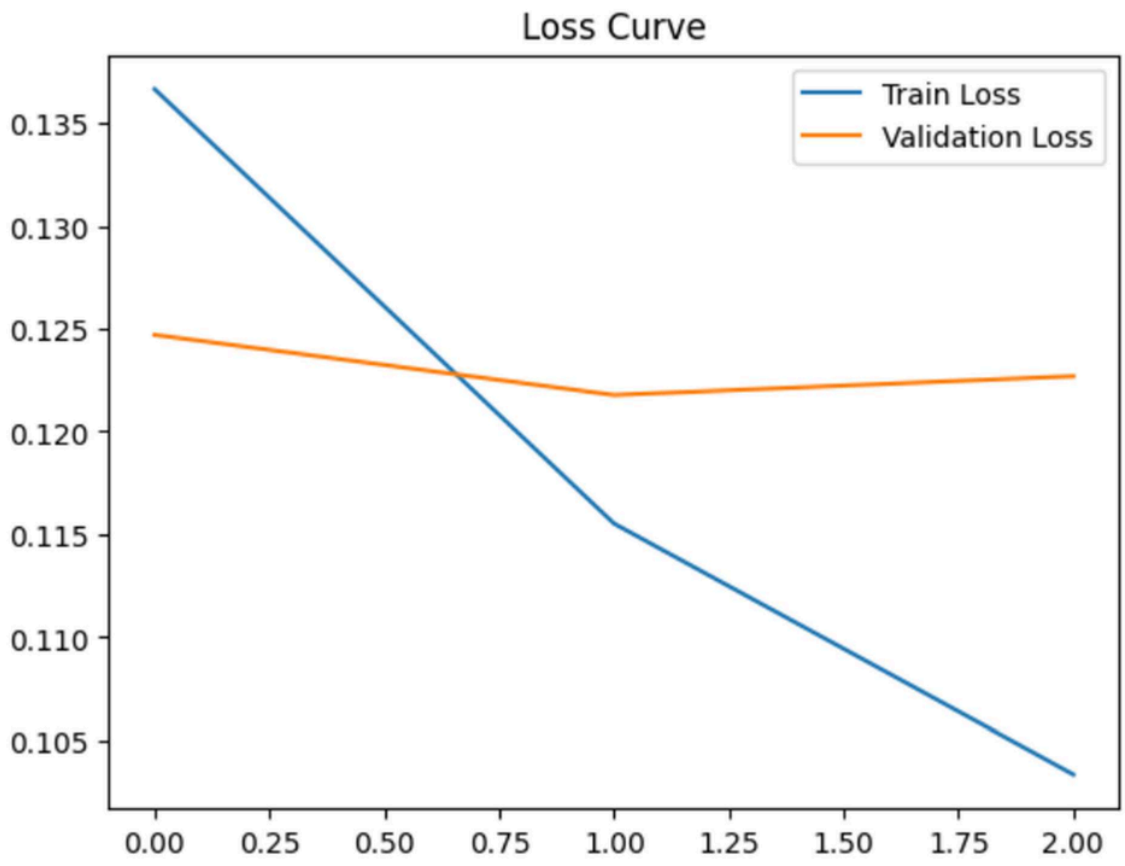
7.3 Confusion Matrix & Graphs

The confusion matrices illustrate the classification performance of the model across different emotion classes, highlighting true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). From the results, it is evident that the model performs well in identifying negative cases (class 0), as seen by the high number of true negatives across all classes (e.g., 3934 in Class 0, 3700 in Class 3). However, the model struggles with correctly predicting positive cases (class 1), particularly in Class 0 where no true positives are observed. This indicates difficulty in detecting certain minority or less frequent emotions. In other classes, such as Class 1 and Class 2, the model shows better balance, with a reasonable number of true positives (225 and 118 respectively), though false negatives are still present.



7.3.1 Confusion Matrix Analysis

The presence of false positives (e.g., 274 in Class 1 and 146 in Class 3) suggests that the model sometimes over-predicts emotions, aligning with the observed higher recall and lower precision. Overall, the confusion matrices confirm that while the model effectively identifies dominant classes, improvements are needed in recognizing minority emotions and reducing misclassifications for better balanced performance.



7.3.2 ROI Loss Curve

The loss curve illustrates the training and validation performance of the model across epochs, providing insights into learning behavior and generalization capability. From the graph, the training loss shows a consistent downward trend, decreasing from approximately 0.137 to 0.103. This indicates that the model is effectively learning patterns from the training data and optimizing its parameters over time. On the other hand, the validation loss initially decreases slightly, from around 0.125 to 0.121, demonstrating that the model is also improving on unseen data during early epochs. However,

after this point, the validation loss begins to stabilize and slightly increase. This divergence between training and validation loss suggests the onset of mild overfitting, where the model continues to improve on training data but does not generalize equally well to new data. Despite this, the gap between training and validation loss remains relatively small, indicating that overfitting is not severe. The model maintains reasonable generalization performance, supported by the balanced evaluation metrics observed earlier. Overall, the loss curve confirms that the training process is stable and effective, with only minor signs of overfitting that can be addressed through techniques such as early stopping or regularization.

7.4 Analysis

The overall training behaviour of the model demonstrates a stable and effective learning process, as evidenced by both the loss curves and evaluation metrics. The training loss shows a consistent and significant decrease across epochs, indicating that the model is successfully learning meaningful patterns from the training data. This steady decline reflects proper optimization of model parameters and suggests that the learning rate, optimizer configuration, and training setup are appropriately chosen. The absence of sudden fluctuations or spikes in training loss further confirms that the model is not experiencing instability during the learning phase. Simultaneously, the validation loss exhibits a slightly different but equally important pattern. Initially, it decreases along with the training loss, showing that the model is improving not only on the training data but also on unseen data. This early improvement indicates that the model is capturing generalizable features rather than simply memorizing the training samples. As training progresses, the validation loss begins to stabilize and shows only a minimal increase.

This behavior suggests that the model reaches an optimal point of generalization early in the training process and maintains that performance across subsequent epochs. The small divergence between training and validation loss indicates mild overfitting, but the gap remains limited, which means the model still generalizes reasonably well.

This pattern of early convergence is particularly important in deep learning models, especially transformer-based architectures. It demonstrates that the model quickly identifies the most relevant linguistic and contextual features required for emotion classification. The stabilization of validation loss implies that additional training epochs contribute only marginal improvements and may even lead to slight overfitting. Therefore, the current training strategy achieves a good balance between learning and generalization.

Further insights can be drawn from the confusion matrices, which provide a class-wise breakdown of model predictions. The matrices reveal that the model performs strongly in identifying the majority classes, as indicated by a high number of true negatives and a reasonable number of true positives. This suggests that the model is effective in distinguishing between the presence and absence of certain emotions. However, some limitations are also evident. In certain classes, particularly those with fewer samples, the number of true positives is relatively low, and false negatives are higher. This indicates that the model struggles to detect less frequent or minority emotions, which is a common challenge in multi-label classification tasks with imbalanced datasets. The presence of false positives in some classes also highlights a trade-off between precision and recall. The model tends to predict more positive labels to ensure that relevant emotions are not missed, which increases recall but slightly reduces precision. This behavior aligns with the observed evaluation metrics and reflects a deliberate balance in the model's prediction strategy. The overall evaluation metrics further support this analysis. The micro precision of approximately 0.42 indicates that a portion of the predicted emotions are incorrect, suggesting room for improvement in reducing false positives. However, the higher micro recall of approximately 0.57 shows that the model is capable of capturing a majority of the actual emotions present in the text. This is particularly important in applications where missing an emotion is more critical than incorrectly predicting one. The micro F1-score of approximately 0.48 represents a balanced combination of precision and recall, demonstrating that the model achieves moderate but stable performance. The difference between micro and macro F1-scores provides additional insight into class-wise performance. While the micro F1-score reflects overall effectiveness across all predictions, the lower macro F1-score indicates that the model does not perform equally well across all emotion classes. This variation is primarily due to class imbalance in the dataset, where frequently occurring emotions are learned more effectively than rare ones. Despite this, the use of weighted Binary Cross-Entropy loss helps mitigate the imbalance to some extent by assigning higher importance to minority classes. Overall, the model demonstrates a well-balanced performance across multiple evaluation dimensions. The combination of stable training loss, controlled validation loss behavior, and consistent evaluation metrics indicates that the system is robust and reliable for multi-label emotion classification. While there are areas for improvement, particularly in handling minority classes and reducing false positives, the current results validate the effectiveness of the proposed approach.

In conclusion, the analysis highlights that the model successfully captures meaningful emotional patterns from textual data and generalizes well to unseen samples. The integration of transformer-based architecture, imbalance-aware optimization, and threshold tuning contributes significantly to this performance. Future improvements could focus on enhancing precision, improving minority class detection, and further reducing overfitting to achieve even better results.

7.5 Summary

The proposed multi-label emotion classification model demonstrates effective and stable learning behavior throughout the training process. The consistent decrease in training loss, along with the stabilization of validation loss, indicates that the model is able to capture meaningful patterns from the data while maintaining good generalization on unseen samples. This balance between learning and generalization is crucial for real-world applications, where the model must perform reliably on new and diverse inputs. The overall performance of the model, reflected by a micro F1-score of approximately 0.48, indicates a moderate yet balanced predictive capability. This score shows that the model effectively manages the trade-off between precision and recall, which is particularly important in multi-label classification tasks. The precision value suggests that while some false positives exist, the model maintains a reasonable level of accuracy in its predictions. At the same time, the higher recall demonstrates the model's strength in identifying a significant portion of relevant emotion labels, ensuring that important emotional cues are not missed. Another key observation is the consistency of performance across different emotion classes. The model performs notably better on well-represented or majority classes such as common emotions, where sufficient training data is available. This is expected, as the model has more

examples to learn from, resulting in stronger and more reliable predictions. However, performance on minority or less frequent classes is comparatively lower, which reflects the inherent challenge of class imbalance in the dataset. Despite this, the use of weighted loss functions and threshold tuning helps improve the detection of these less frequent emotions to some extent.

The results also highlight the effectiveness of transformer-based architectures in handling complex natural language tasks. The model successfully captures contextual and semantic relationships within text, enabling it to detect multiple emotions simultaneously. This capability is essential for real-world scenarios, where human emotions are often overlapping and not limited to a single category.

Overall, the model provides a strong baseline for multi-label emotion classification. It demonstrates reliable and consistent performance, making it suitable for practical applications such as sentiment analysis, social media monitoring, and conversational systems. While there is room for improvement—particularly in enhancing precision and minority class detection— the current system establishes a solid foundation for future advancements and optimizations in emotion-aware natural language processing.

CHAPTER 8

Conclusion

This project presents a comprehensive approach to multi-label emotion detection from textual data using a transformer-based deep learning model. The primary objective was to develop an efficient and reliable system capable of identifying multiple emotions within a single piece of text, addressing the limitations of traditional sentiment analysis methods. By leveraging advanced natural language processing techniques and the power of contextual embeddings, the proposed system successfully demonstrates the effectiveness of modern transformer architectures in handling complex emotion classification tasks. The use of the RoBERTa model plays a central role in the system's success. Its ability to capture deep contextual relationships through self-attention mechanisms allows it to understand subtle emotional cues, semantic nuances, and dependencies within text. Unlike earlier models such as Support Vector Machines, Naïve Bayes, or even basic deep learning approaches like CNNs and RNNs, RoBERTa provides a more robust and scalable solution for emotion detection. This is particularly important in multi-label classification scenarios, where multiple emotions can coexist and interact within the same sentence. A significant challenge addressed in this work is the issue of class imbalance, which is common in real-world emotion datasets such as GoEmotions. Certain emotions appear far more
frequently than others, making it difficult for models to learn and predict minority classes effectively. To overcome this, the project incorporates a weighted Binary Cross-Entropy loss function, which assigns greater importance to underrepresented classes. This approach improves the model's ability to detect less frequent emotions and contributes to a more balanced
overall performance.

Another key contribution of this work is the implementation of adaptive threshold tuning. Instead of relying on a fixed threshold for converting probability outputs into binary predictions, the system evaluates multiple threshold values and selects the one that maximizes the F1-score.

This strategy enhances the balance between precision and recall, leading to more accurate and meaningful predictions. It also reflects a practical understanding of multi-label classification, where optimal thresholds may vary depending on the dataset and model behavior.

The experimental results demonstrate that the model achieves moderate yet stable performance across various evaluation metrics. With a micro F1-score of approximately 0.48, precision around 0.42, and recall around 0.57, the system effectively balances the trade-off between identifying relevant emotions and minimizing incorrect predictions. The higher recall indicates that the model is capable of capturing most of the true emotions present in the text, which is particularly valuable in applications where missing emotional information can have significant consequences. At the same time, the moderate precision suggests that there is still room for improvement in reducing false positives.

The analysis of confusion matrices further supports these findings, showing that the model performs well on majority classes while facing challenges with minority and less frequent emotions. This behavior is expected given the nature of the dataset and highlights the importance of continued research in handling class imbalance more effectively. Additionally, the loss curve analysis confirms that the model undergoes stable training, with a clear convergence pattern and only mild signs of overfitting. This indicates that the chosen training strategy, including the use of the Adam optimizer and regularization techniques, is effective in achieving reliable performance.

Despite its strengths, the system has certain limitations. The reliance on a large transformer model such as RoBERTa-large results in high computational requirements, making training time-consuming and resource-intensive. Furthermore, the model may struggle with highly ambiguous, sarcastic, or context-dependent expressions, which remain challenging even for advanced NLP systems. These limitations highlight potential areas for future improvement. Future work can focus on enhancing the efficiency and scalability of the model by exploring lighter architectures such as RoBERTa-base or DistilBERT. Additionally, incorporating techniques such as attention refinement, data augmentation, or ensemble learning could further improve performance, particularly for minority classes. Expanding the system to support multilingual emotion detection and real-time deployment would also increase its applicability in diverse real-world scenarios.

In conclusion, this project successfully demonstrates the potential of transformer-based models for multi-label emotion detection. By integrating contextual feature extraction, imbalance-aware optimization, and adaptive thresholding, the proposed system provides a robust and scalable solution for understanding human emotions in text. The results validate the

effectiveness of the approach and establish a strong foundation for future advancements in emotion-aware natural language processing systems.

CHAPTER 9

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